4. Write a Python program to demonstrate the confidentiality property a VPN tunnel provides, showing that data captured in transit is unreadable without the shared key, using a simple symmetric stream cipher as a stand-in for a real VPN's encryption.

[6] ✓ 17s

```
def encrypt_decrypt(data, key):
    result = []

    for i in range(len(data)):
        value = ord(data[i]) ^ ord(key[i % len(key)])
        result.append(value)

    return result


message = input("Enter message: ")
key = input("Enter shared key: ")

if key == "":
    print("Key cannot be empty")

else:
    encrypted = encrypt_decrypt(message, key)

    print("\nEncrypted ciphertext (hex):",
          " ".join(format(x, "02x") for x in encrypted))

    print("Captured data is ciphertext, not readable plaintext.")
```

[6] ✓ 17s

```
    decrypted = ""

    for i in range(len(encrypted)):
        value = encrypted[i] ^ ord(key[i % len(key)])
        decrypted += chr(value)

    print("Decrypted message:", decrypted)

    wrong_key = "wrongkey"
    wrong_decryption = ""

    for i in range(len(encrypted)):
        value = encrypted[i] ^ ord(wrong_key[i % len(wrong_key)])
        wrong_decryption += chr(value)

    print("Decryption with incorrect key:",
          repr(wrong_decryption))
```

```
Enter message: HELLO
Enter shared key: KEY

Encrypted ciphertext (hex): 03 00 15 07 0a
Captured data is ciphertext, not readable plaintext.
Decrypted message: HELLO
Decryption with incorrect key: 'trzim'
```